

Aviation Accidents & Incidents

حوادث ووقائع الطيران: NTSB, FAA, وخدمات الطيران العالمية

An In-Depth Comparative Study of Safety Oversight, Forensic Investigation & Global Frameworks

دراسة شاملة ودقيقة للرقابة على السلامة والتحقيقات الجنائية والهيئات العالمية



Key Regulators & Global Bodies

Understanding the Pillars of Aviation Safety and Investigation

فهم الركائز الأساسية لسلامة الطيران والتحقيق في الحوادث

Aviation Safety Fundamentals

Defining Accidents vs. Incidents

Accident (حادث): An occurrence associated with the operation of an aircraft resulting in fatal injury, illness, or structural damage.

Incident (واقعة): An event, other than an accident, affecting or affecting the safety of operation.

Primary Goal: Modern aviation safety focuses on proactive risk reduction rather than assigning civil or criminal liability.

الهدف الأساسي هو تحديد الأسباب الجوهرية لمنع تكرار الكوارث مستقبلاً.



NTSB vs FAA: Key Roles

NTSB (المجلس الوطني لسلامة النقل)

Independent Investigator: Lead agency in investigating civil aviation accidents in the U.S.

No Regulatory Power: Cannot issue fines or binding laws; focuses solely on determining probable cause and safety recommendations.

جهة مستقلة تماماً تركز على اكتشاف الأسباب المحتملة وتوصيات السلامة.

FAA (إدارة الطيران الفيدرالية)

Regulatory & Operational Body: Manages air traffic control, certifies pilots, and sets airworthiness standards.

Enforcement Mandate: Can issue Airworthiness Directives (ADs), impose fines, or revoke certifications.

سلطة تنظيمية وتنفيذية تفرض القوانين وتدير المجال الجوي.

NTSB: Independent Inquiries

The "Go Team" Deployment

Upon notification of a major crash, the NTSB deploys a multi-disciplinary "Go Team" consisting of experts in systems, engines, structures, and human performance.

Party System: Invites manufacturers and operators to provide technical assistance under strict NTSB supervision.

يقود فريق النشاط السريع "Go Team" التحقيقات الميدانية بدقة متناهية.



FAA Operations & Oversight

Air Traffic Control

Manages over 45,000 daily flights across the National Airspace System ensuring safe separation.

إدارة وتوجيه الحركة الجوية عبر المجال الجوي.



Aircraft Certification

Sets rigorous design and airworthiness standards for manufacturers before flight operations.

إصدار شهادات الصلاحية وترخيص الطائرات.



Rulemaking Action

Enforces safety recommendations issued by the NTSB into legally binding mandatory rules.

تحويل التوصيات إلى قوانين إلزامية.

ICAO & Global Aviation Services



ICAO Annex 13 Framework

Sets international protocols for crash investigation protocols worldwide.

الملحق 13 لمنظمة الطيران المدني الدولي.



Global Airline Standards

Harmonizing pilot training, crew resource management (CRM), and safety checks.

توحيد معايير التدريب والسلامة العالمية.



Cross-Border Support

Coordination between foreign authorities (e.g., BEA, AAIB) and NTSB/FAA.

التعاون الدولي بين هيئات التحقيق المعتمدة.

Aviation Safety Key Metrics

99.9%

Safety Reliability Index

معدل موثوقية وأمان رحلات الطيران التجاري

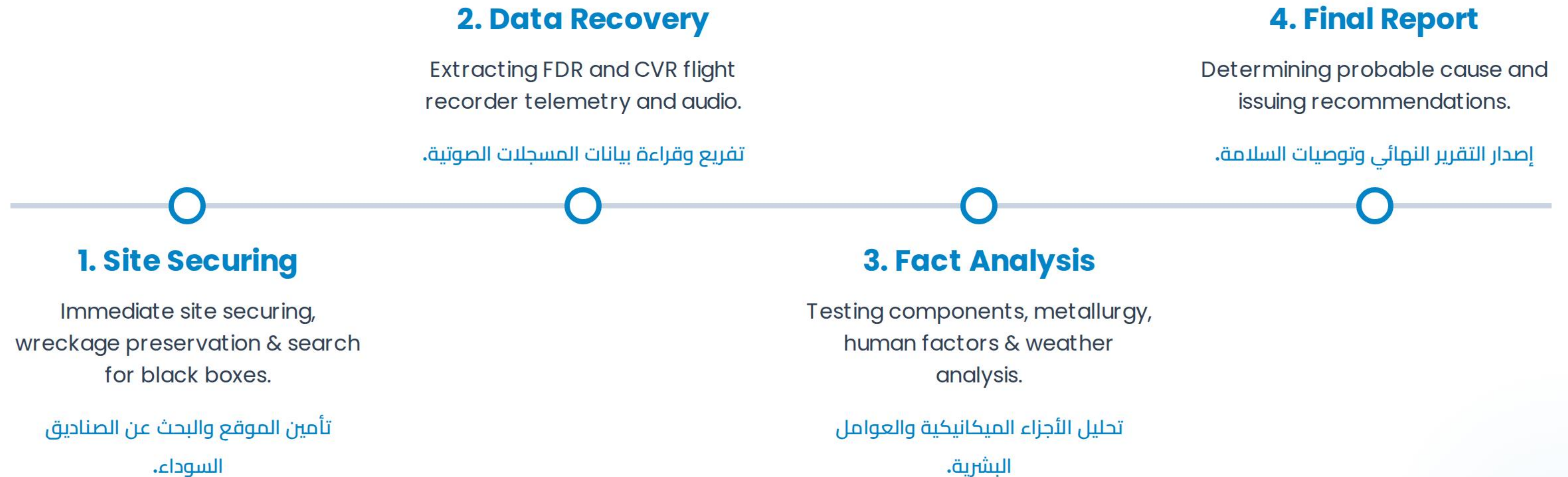
Continuous Improvements in Sky Safety

Commercial aviation remains the safest mode of long-distance transportation worldwide.

Due to rigorous NTSB inquiries and FAA mandates, commercial airliner fatal accident rates have dropped by over 80% over the last four decades.

انخفاض معدل الحوادث المميتة بنسبة تتجاوز 80% بفضل التحقيقات والتحديثات التنظيمية المستمرة.

Accident Investigation Steps



Accident Severity Classes

Class Level	Investigation Focus	Resources Assigned	Primary Objective / Goal
Class 1 (الفئة الأولى)	Major transport airliner crashes with fatal casualties.	Full NTSB Go Team + Industry Parties.	In-depth probable cause & systemic safety fixes.
Class 2 (الفئة الثانية)	Significant accidents involving notable damage/injuries.	Senior Regional Investigator led team.	Identify safety factors & recurring failure trends.
Class 3 (الفئة الثالثة)	Incidents revealing specific underlying risk issues.	Specialized Subject Matter Experts.	Develop preventative industry safety messages.
Class 4 (الفئة الرابعة)	Limited-scope general aviation incidents.	Single Investigator (mostly remote).	Quick factual reporting within 6 months.

Flight Recorders & Forensics

-  **Flight Data Recorder (FDR) / مسجل بيانات الطيران:** Captures hundreds of parameters including altitude, airspeed, heading, engine thrust, and control surface positions.
-  **Cockpit Voice Recorder (CVR) / مسجل الأصوات بقمرة القيادة:** Records cockpit conversations, radio communications, and audible alarm sounds.
-  **Metallurgical & Forensic Analysis / التحليل الجنائي الميكانيكي:** Examines structural fractures under electron microscopes to identify fatigue cracks versus impact failure.
-  **Real-time Telemetry (FDM) / النقل اللحظي للبيانات:** Modern trends shifting towards satellite streaming of data during critical flight phases.



Questions & Discussion

أسئلة ومناقشة

Thank you for your attention to Aviation Safety & Investigation Governance

شكراً لاهتمامكم بسلامة الطيران وهيئات التحقيق والمتابعة العالمية

NTSB & FAA Safety Standards | Global Aviation Service Framework

Image Sources



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